

3. Find all the IP addresses on your network. Unicast, Multicast, and Broadcast on your network Aim

To identify the IP addresses on a local network and understand the concepts of **Unicast, Multicast, and Broadcast** addressing.

Identifying Local IP Addresses

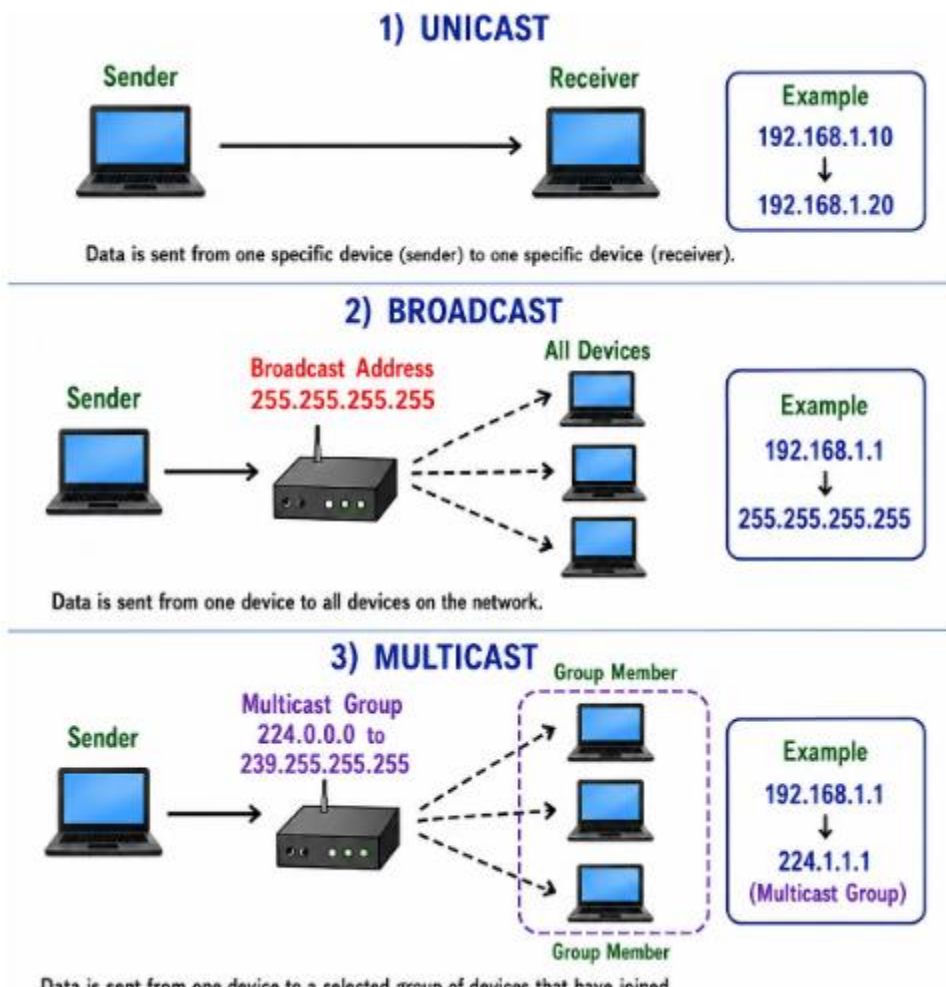
- **Command Line Tools:** Run `arp -a` in your terminal or command prompt to see a list of IP addresses and MAC addresses of devices active on your local network segment.
 - **Network Scanners:** Use software tools like Angry IP Scanner or nmap (e.g., `nmap 192.168.1.0/24`) to scan the local subnet range and discover active hosts.
 - **Router Admin Page:** Log into your Wi-Fi router's web interface to view the DHCP client table showing all connected devices and their assigned local IP addresses.
-
- **Unicast Addresses**
 - One-to-one communication sent to a single specific device.
 - Range: Any valid host IP assigned dynamically via DHCP or statically within your local subnet (e.g., 192.168.x.x or 10.x.x.x).
 - Default for standard web browsing, file transfers, and individual device traffic.

Multicast Addresses

- One-to-many communication sent to a subscribed group of devices.
- Range: 224.0.0.0 to 239.255.255.255 (IPv4 Class D).
- Common local group example: 224.0.0.251 (Multicast DNS) or streaming media protocols.

Broadcast Addresses

- One-to-all communication sent to every node on the local subnet.
- Limited Broadcast: 255.255.255.255 (reaches all nodes on the immediate local network).
- Directed Broadcast: The final address of your specific subnet range (e.g., 192.168.1.255 for a 192.168.1.0/24 network).



Result

The IP addresses available on the local network were successfully identified and analyzed.

4. Use Packet tracer software to build network topology and configure using Distance vector routing protocol.

Aim

To build a network topology in Cisco Packet Tracer and configure a **Distance Vector Routing Protocol (RIP)** to enable communication between different networks.

Procedure

Configuration of network topology using distance vector routing protocol:

Steps to Configure and Verify Three Router Connections in Cisco Packet Tracer using RIP Routing:

Step 1: First, open the Cisco packet tracer desktop and select the devices given below:

S No	Devices	Model Name	Qty
1	PC	PC	6
2	Switch	PT-Switch	3
3	Router	PT-Router	3

Routing Table:

Routing Table:

S No	Devices	IPv4 Address	Subnet mask	Default Gateway
1	PC0	192.168.10.2	255.255.255.0	192.168.10.1
2	PC1	192.168.10.3	255.255.255.0	192.168.10.1
3	PC2	192.168.20.2	255.255.255.0	192.168.20.1
4	PC3	192.168.20.3	255.255.255.0	192.168.20.1

5	PC4	192.168.30.2	255.255.255.0	192.168.30.1
6	PC5	192.168.30.3	255.255.255.0	192.168.30.1

Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
 - Then, go to desktop and then IP configuration and there you will IPv4 configuration.
 - Fill IPv4 address and subnet mask.
-
- Assigning an IP address using the ipconfig command, or we can also assign an IP address with the help of a command.
 - Go to the command terminal of the PC.
 - Then, type ipconfig <IPv4 address><subnet mask><default gateway>(if needed)

Example: `ipConfig 192.168.10.2 255.255.255.0 192.168.10.1`

Repeat the same procedure with other PCs to configure them thoroughly.

Step 3: Configure router with IP address and Subnet mask.

IP Addressing Table Router:

S No	Device	Interface	IPv4 Address	Subnet mask
1	Router 0	fastEthernet0/0	192.168.10.1	255.255.255.0
		Serial 2/0	10.0.0.1	255.0.0.0
2	Router 1	FastEthernet0/0	192.168.20.1	255.255.255.0
		Serial 2/0	20.0.0.1	255.0.0.0
3	Router 2	FastEthernet0/0	192.168.30.1	255.255.255.0
		Serial 2/0	30.0.0.1	255.0.0.0

- To assign an IP address in router0, click on router0.
 - Then, go to config and then Interfaces.
 - Make sure to turn on the ports.
 - Then, configure the IP address in FastEthernet and serial ports according to IP addressing Table.
 - Fill IPv4 address and subnet mask.
-
- Repeat the same procedure with other routers to configure them thoroughly.

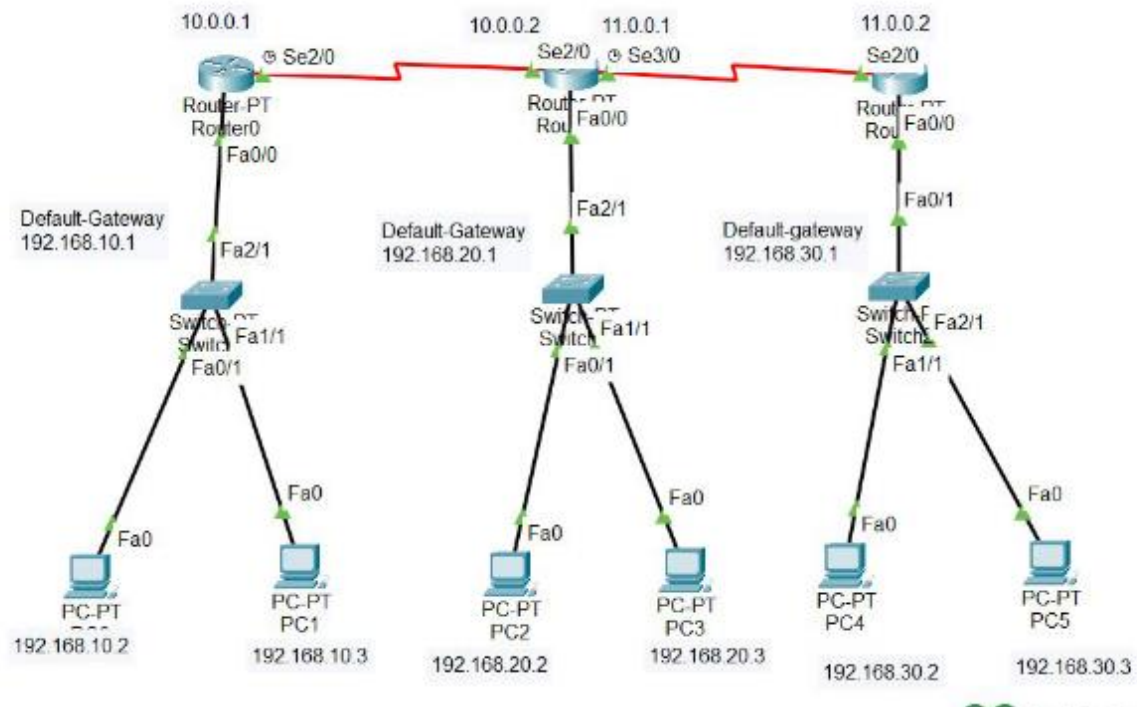
Step 4: After configuring all of the devices we need to assign the routes to the routers.

Step 5: Verifying the network by pinging the IP address of any PC.

- We will use the ping command to do so.
- First, click on PC0 then Go to the command prompt.
- Then type ping <IP address of targeted node>.
- As we can see in the below image we are getting replies which means the connection is working properly.

Example : ping 192.168.20.2

A simulation of the experiment is given below we are sending PDU from PC0 to PC2 and PC3 to PC5



Result

The network topology was successfully created in Cisco Packet Tracer, and the **Distance Vector Routing Protocol (RIP)** was configured on all routers. The routers exchanged routing information successfully, and communication between devices on different networks was verified .